Chapter 3 Teacher Key — Energy and Speed Are Related

> Combined answer keys: Ramp It Up! 2.0 · Drop It · Practice · Chapter 3 Quiz.

Answer Key — Ramp It Up! 2.0

- **Artifact:** `production/worksheets/03-ch3-ramp-it-up-student.docx`
- **Source alignment:** AP 4.1 (Ramp It Up, Day 1); Reader Ch.3 pp. 13–16; TG L4. ADAPTED (3 heights + trials + cm + graph).

Item	Correct/expected response	Acceptable alternatives	Reasoning	Common misconception	Scoring note	What NOT to penalize
PREDICT	a guess (e.g., "3 books moves the cup farthest")	any prediction	prediction = testable guess	wrong prediction = failure	1 pt	a "wrong" prediction
FAIR TEST (change/keep/measure)	change ramp height; keep ball/cup/surface/release point/method; measure cup displacement (cm)	paraphrases	fair-test isolation of one variable	thinking speed is directly measured	3 pts (1 each)	informal phrasing
DATA TABLE	displacement increases with ramp height; averages computed correctly	any internally consistent data	repeated trials + average	recording "speed" in a data cell	6 pts (table complete + averages)	noisy/ uneven trial data
GRAPH	title "Ramp Height vs. Average Cup Displacement"; x = books, y = cm; 3 bars/points rising	any correct axes/plot	graph shows the pattern	labeling it "Speed vs. Energy"	4 pts	bar vs. dot style
OBSERVE	higher ramp → farther cup	"the cup moved more at 3 books"	pattern in data	—	1 pt	—
INFER	faster motion of the same ball is associated with greater energy of motion	"higher ramp → faster → more energy (inferred)"	inference from measured change	claiming we measured energy	2 pts	informal phrasing
EXPLAIN (CER)	Claim + evidence (measured	any claim + measured	change = evidence of	stating energy as directly observed	3 pts	full formal CER not

Item	Correct/expected response	Acceptable alternatives	Reasoning	Common misconception	Scoring note	What NOT to penalize
	displacement) + reasoning (greater change → greater energy of motion, inferred)	change + reasoning	energy			required

Notes

- Students must NOT claim they measured speed or energy. If they write "we measured the speed," do not award the INFER/EXPLAIN inference credit — redirect to "we measured the change."
- The "faster" determination is an INFERENCE from ramp height, not a measurement.
- A wrong prediction is NOT penalized if the observation/explanation are correct.

Answer Key — Drop It

- **Artifact:** `production/worksheets/03-ch3-drop-it-student.docx`
- **Source alignment:** AP 4.2 (Drop It, Day 2); Reader Ch.3 pp. 13–16; TG L4. ADAPTED (fixed cm heights + cm crater width; depth qualitative).

Item	Correct/expected response	Acceptable alternatives	Reasoning	Common misconception	Scoring note	What NOT to penalize
PREDICT	a guess (e.g., "90 cm makes the biggest crater")	any prediction	prediction = testable guess	wrong prediction = failure	1 pt	a "wrong" prediction
FAIR TEST (change/keep/measure)	change drop height; keep ball/sand tray/surface/drop technique; measure crater width (cm)	paraphrases	fair-test isolation	thinking height = speed (it's a cause, speed is inferred)	3 pts	informal phrasing
DATA TABLE	crater width increases with drop height; averages computed	any internally consistent data	repeated trials + average	recording "energy" in a data cell	6 pts	noisy crater data

Item	Correct/expected response	Acceptable alternatives	Reasoning	Common misconception	Scoring note	What NOT to penalize
OBSERVATION	60 cm (or 90 cm) caused the greatest change; depth note (medium/deep)	any correct height + qualitative depth	higher → faster → bigger crater	—	2 pts	depth wording
EXPLANATION	higher drop → faster fall (inferred) → bigger crater (measured) → greater energy of motion (inferred)	claim + measured change + inference	change = evidence of energy	claiming we measured energy or speed	3 pts	informal phrasing

Notes

- Crater depth is qualitative only (shallow / medium / deep) — do not require millimeter precision.
- Students must NOT claim they measured the ball's speed or energy. The "faster" determination is an inference from drop height.
- Fixed heights (30/60/90 cm) replace the official knee/waist heights for a fairer test (see `curriculum/chapter-03/source-map.md`).

Answer Key — Practice (Energy and Speed Are Related)

- **Artifact:** `production/worksheets/03-ch3-energy-speed-practice.docx`
- **Source alignment:** Reader Ch.3; TG L4/L5. Sections CUSTOM/ADAPTED.

Section	Correct/expected response	Acceptable alternatives	Reasoning	Common misconception	Scoring note	What NOT to penalize
A. Same object or not?	1 SAME · 2 NOT · 3 SAME · 4 NOT	correct classification	only same-object comparisons isolate speed	"more mass = more energy" is NOT the target	4 pts (1 each)	—
B. Read the data	fastest = 3 books (highest ramp); "we did not measure speed — we infer faster from the higher ramp"	"3 books" + inference	ramp height → relative speed is inferred	"the ruler measured speed"	3 pts	informal phrasing
C. Read the graph	taller bar at 3 books → faster ball caused greater change → greater energy of	claim + change	graph shows the pattern	—	2 pts	—

Section	Correct/expected response	Acceptable alternatives	Reasoning	Common misconception	Scoring note	What NOT to penalize
	motion (same ball)					
D. Claim + evidence	circle 1 and 3	correct evidence selections	evidence = observed change of the SAME object	selecting the mass item (#4) or the color item (#2)	2 pts	—
E. Fix the scientist	the object changed too (heavier vs lighter), so it is not a same-object comparison — speed was not isolated	identifies the confounded variable (mass)	fair-test logic	—	2 pts	—

Notes

- Section D item 4 ("a heavier ball moved the cup farther") is the confounded-variable distractor — it must NOT be selected as evidence for the speed↔energy relationship.
- Section E requires identifying that MASS changed, not speed; do not award "speed must be the reason."

Answer Key — Chapter 3 Quiz

- **Artifact:** `production/quizzes/03-ch3-energy-speed-quiz.docx`
- **Source alignment:** Reader Ch.3 pp. 13–16; TG L4/L5; AP 4.1/4.2. PE 4-PS3-1.

Item	Correct/expected response	Acceptable alternatives	Reasoning	Common misconception	Scoring note	What NOT to penalize
1	a) a measurement of the distance an object travels over an amount of time	—	Reader p.13	speed = "how fast" (colloquial)	1 pt	—
2	b) energy of motion	—	Reader p.9 (Ch.2)	energy of motion = "sound energy"	1 pt	—
3	b) the SAME ball moving slowly, then faster	—	TG L5 Know the Science (same-object comparison)	picking a mass comparison (a, d)	1 pt	—
4	greatest change = 3 books; the group	correct height + "measured the	measured vs	"we measured the	3 pts	informal

Item	Correct/expected response	Acceptable alternatives	Reasoning	Common misconception	Scoring note	What NOT to penalize
	measured the change (cup displacement), NOT the speed	change"	inferred	speed"		phrasing
5	taller bar at higher ramp → faster → greater change → greater energy of motion (same ball)	claim + change	graph interpretation	"the graph measured energy"	2 pts	—
6	M: cup moved 18 cm · ramp height 3 books; I: ball faster · faster ball more energy	correct classification (2 M, 2 I)	measured vs inferred	treating inference as measurement	4 pts (1 each)	—
7	hard throw = greater energy of motion; evidence = the hand moved back + sting (greater change)	claim + change	Reader p.14	—	2 pts	informal phrasing
8	the ruler measured the change (displacement), not energy; energy is inferred	identifies the error	TG L4 Know the Standards	—	2 pts	—
9	c) greater	—	Reader p.14	—	1 pt	—
10 (BONUS)	keeping the object the same makes sure only speed changes (mass/type don't confound the comparison)	fair-test reasoning	TG L5 Know the Science	—	1 pt bonus	—

Out-of-scope confirmation

- No mass comparison as central · no collisions · no transfer/transformation · no engineering · no kinetic/potential · no "we measured energy."

Total

~18 points + 1 bonus (adjust denominator per school grading policy — see OPEN_QUESTIONS).